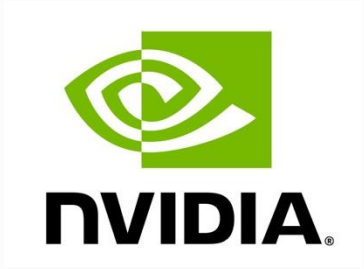
Accelerate HPC and AI workloads with the NVIDIA GH200 Superchip and HPE EX Supercomputing Platform.

Richard Gilham, Bristol Centre for Supercomputing

Filippo Spiga and Karin Sevegnani, NVIDIA

Jess Jones and Tim Dykes, HPE









Housekeeping

Fire alarms- not aware of a test. Follow instructions and signage

Comfort- take breaks when you need. Follow signage. Try not to disrupt others.

User accounts

Those pre-registered with acceptable institutional email addresses will have received invitations

- There are strict compliance criteria we must follow

Please sign in with Richard, otherwise accounts will be disabled

Do not abuse the compute- there are finite GPU hours!

Timetable

0930-1030 Richard Gilham

Welcome & Logistics. Introducing Isambard-AI

Practical- Finding your way around Isambard-AI

1030-1100: Tim Dykes (HPE)

HPE Cray EX4000 platform + NVIDIA 4-way GH200 Superchip: Hardware and Software Deep Dive

1100-1130 Coffee Break

1130-1300 Filippo Spiga (NVIDIA)

NVIDIA GH200 Programming Models Deep Dive

Practical- Guided hands-on session / “Bring-Your-Own-Code” [single node / single GPU]

1300-1400 Lunch Break

1400-1530 Jess Jones (HPE)

Tips & tricks for multi-node scaling AI workloads on EX4000 and HPE Slingshot

Practical- Guided hands-on session / “Bring-Your-Own-Code” [multi node / multi GPU]

1530-1600 Coffee Break

1600-1730 Karin Sevegnani (NVIDIA)

Profiling Large Language Model Trainings on the Grace Hopper Superchip using Nsight Systems

Guided hands-on session / “Bring-Your-Own-Code” [profiling and performance analysis]

Introducing Isambard-AI

Richard Gilham

Bristol Centre for Supercomputing

richard.gilham@bristol.ac.uk | https://docs.isambard.ac.uk

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BriCS Bristol Centre for Supercomputing

Isambard-AI Origins & Timeline

Government feedback on Bristol Isambard-AI proposal



AI Safety Summit



First Modular Data Centre installation



Onboarding of early tech prep users



Isambard-AI Phase 2 installation

Summer | Autumn | Winter | Spring | Summer | Autumn | Winter | Spring | Summer | Autumn

2023 | 2024 | 2025

Isambard-AI procurement

Isambard team recruitment



Isambard-AI Phase 1 power-on



First DSIT allocated projects

Isambard-AI Phase 2 acceptance

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May 2024


April 2025

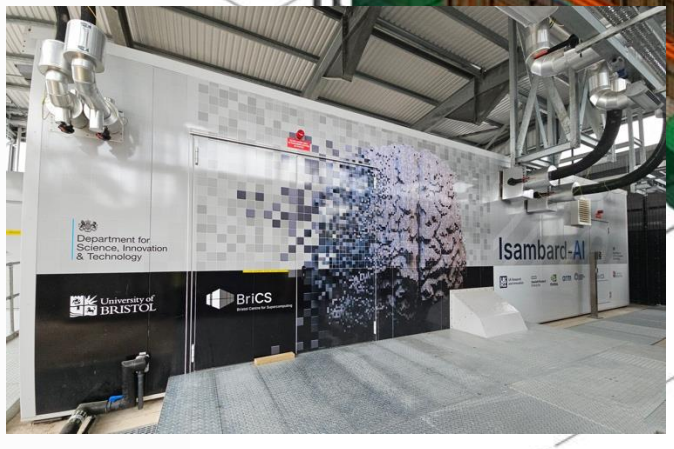
Isambard Park 1



Heat Recovery

Power

Isambard-AI Phase 1
Isambard 3



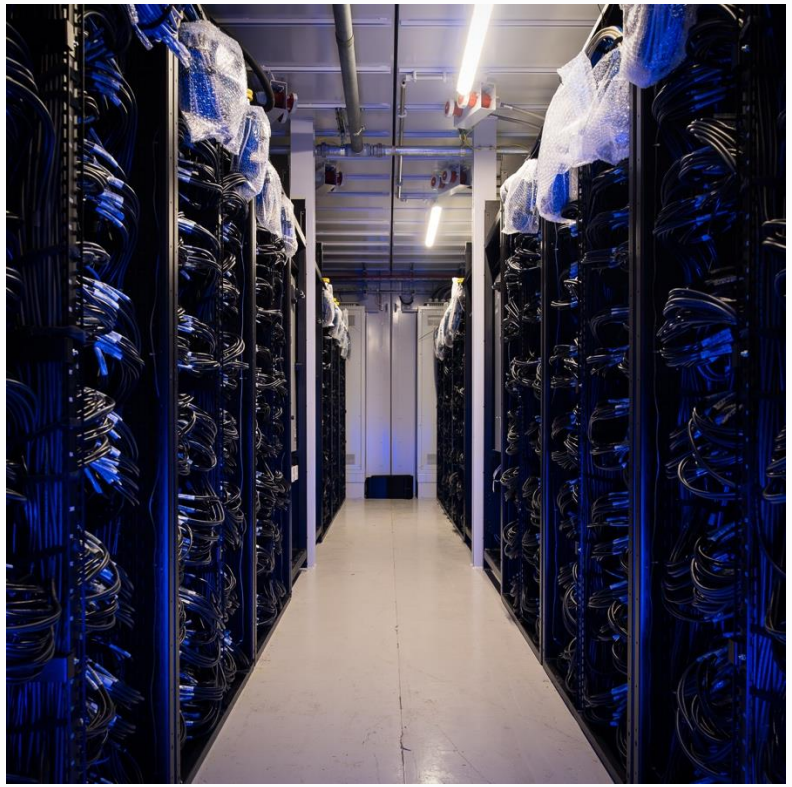


Isambard-AI Phase 2

Cooling

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World-class compute

Isambard-AI phase 2:

- #11 Top500 (June-25)
- About 20,000 laptops fast for AI

11	**Isambard-AI phase 2** - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS, HPE University of Bristol United Kingdom	

Isambard-AI phase 1:

- #128 Top500 (June-24)
- #2 Green500 (June-24)

Laptop+NVIDIA GeForce RTX 4070
= 466 AI TOPS
Isambard-AI Phase 2: GH200 x 5280
= 10.4M AI TOPS





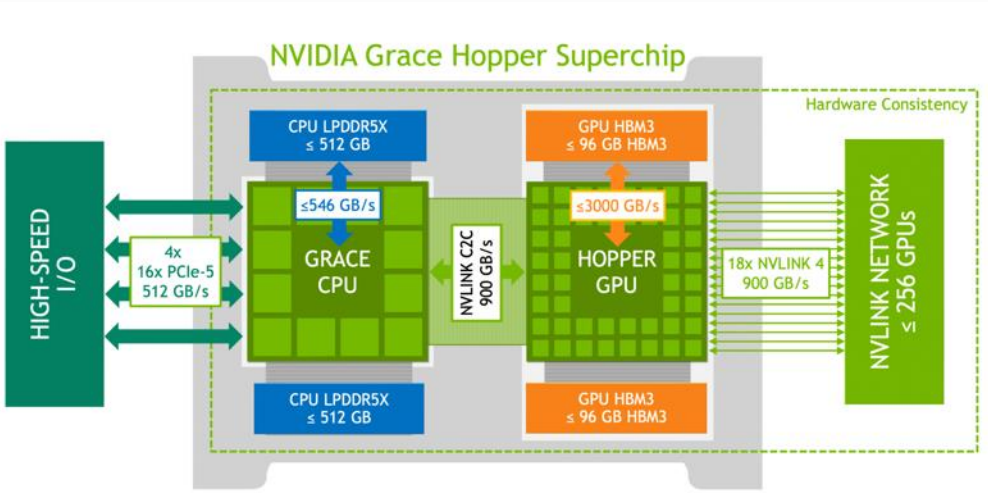
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TOPS = measure of AI performance in trillions(!) of operations per second

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Isambard-AI

	Phase 1	Phase 2
Nodes	42	1320
GH200 superchips (4/node)	168	5,280
Grace CPU cores (4x72/node)	12,096	380,160
GPUs (4/node)	**168**	**5,280**
LPDDR5 CPU memory (4x120GB/node)	21.5 TB	675 TB
HBM3 GPU memory (4x96GB/node)	16.1 TB	506 TB
Total memory	37.6 TB	1.18 PB
Storage	~1 PB all-flash	All-flash ~20 PB POSIX ~7 PB object



Source: NVIDIA Grace Hopper Superchip Architecture Whitepaper

Intra-node connectivity:

NVLink-C2C (900 GB/s)

Interconnect:

4 x Slingshot 11 (200 Gbps)

https://docs.isambard.ac.uk/specs/

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Isambard 3



2023- £10M investment

Traditional Tier-2 HPC: mostly CPU + batch

Modular Data Centre: MO data centre unavailable

384 nodes NVIDIA Grace CPU Superchips:

- 2x72 cores; 2x120 GB LPDDR5, slingshot 11 interconnect

Multi-Architecture Comparison System (MACS):

- AMD Milan/Genoa/Bergamo; Intel SPR DDR5/HBM
- AMD Milan + GPUs: AMD MI100, NVIDIA A100/H100 PCIe)

Access and User Experience

Services and User Experience

Cloud-like service provision

Give users agency (and responsibility)

Federation and democratisation

- New sectors (HPC users != AI users)
- Easy to port across DRI

Quick access to AI supercomputing

https://portal-airr.isambard.ac.uk
- Allocators run access calls
- Prospective Pis apply

https://portal.isambard.ac.uk
- PIs receive invitation and resource allocation
- Accept T&Cs
- Invites & manages researchers
- Monitor resource usage

MyAccessID

Access compute
- SSH
- JupyterHub

Support
- https://support.isambard.ac.uk
- https://docs.isambard.ac.uk

IRR Gateway route
10,000 GPU hours, designed for researchers from academia, industry, or other UK organisations

AIRR Rapid Access route
20,000 GPU hours, is open to UK-registered micro, small and medium organisations

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Practical- Finding your way around Isambard-AI

Aims

Log in and understand the role of clifton

Complete tasks with the help of the documentation

- Using the module system
- Using the Slurm workload manager
- Build and run simple examples

https://docs.isambard.ac.uk/training/ieee_cluster2025/

https://github.com/isambard-sc/ieee_cluster2025

```
ssh t5c.aip2.isambard

--reservation=IEEE_Cluster_Tutorial
```

Do not run long/large jobs… there are finite GPU hours!